k-large-prime ladder: multi-target batch and triple large primes

Protocol `PROTOCOL.md`, frozen before data, with Amendments 1 and 2 (also frozen before their data). Every run's answer checked against the planted log; 0 failures in all cells. Goal (index calculus near rho parity) **not met**.

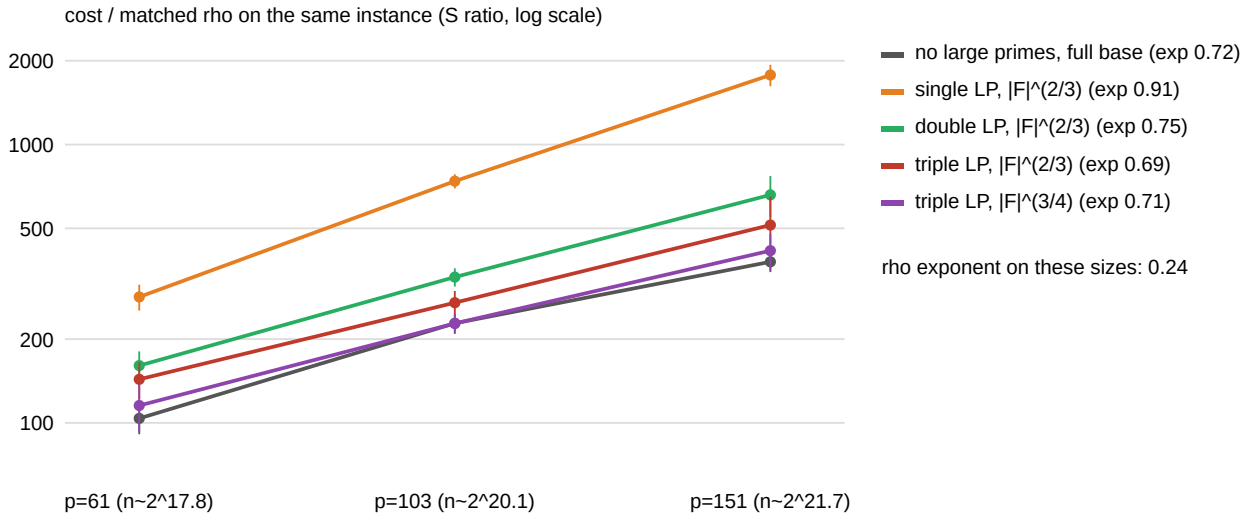
Amendment 1: multi-target batch (k = 1), prime-field curves 18/20/22 bits, 16 trials per cell

bits	T	B=0 cost / sqrt r per target	B=1 / B=0	gain vs T=1 (B=0)
18	1	1.922	1.278 ± 0.176	1.00
18	4	0.971	1.039 ± 0.094	1.98
18	16	0.615	0.945 ± 0.044	3.12
18	64	0.375	1.062 ± 0.028	5.12
20	1	1.846	1.119 ± 0.134	1.00
20	4	0.817	1.108 ± 0.100	2.26
20	16	0.535	1.033 ± 0.063	3.45
20	64	0.348	0.949 ± 0.039	5.30
22	1	1.580	1.314 ± 0.202	1.00
22	4	0.793	1.193 ± 0.081	1.99
22	16	0.502	1.046 ± 0.061	3.15
22	64	0.325	0.966 ± 0.051	4.86

Sharing one table across T = 64 targets cuts per-target cost 4.9-5.3x (ideal sqrt(64) = 8x). The residual collector's factor-base unknown (B = 1) costs 1.12-1.31x at T = 1 and 0.95-1.06x at T = 64: in a batch it reaches parity with multi-target rho (B = 0) and does not beat it. Comparisons are to multi-target rho at the same T, never to single-target rho.

Amendment 2: triple large primes, existing Gaudry engine on E/F_(p^3), 8 instances per size

Setting differs from the main protocol: a cubic extension field, where a subfield factor base exists (the prime-field k = 2..4 script could not be written in this session). Relations are 3-point decompositions, so quadruple large primes are not expressible in this engine. Cost is the engine's own S (field multiplications converted to additions), ratio to rho on the same instance.



arm	p=61	p=103	p=151	exponent (log2 ops on log2 n)
no large primes, full base	104 ± 13	228 ± 16	379 ± 31	0.721
single LP, |F|^(2/3)	283 ± 30	739 ± 43	1778 ± 158	0.908
double LP, |F|^(2/3)	161 ± 20	334 ± 25	660 ± 111	0.750
triple LP, |F|^(2/3)	143 ± 22	270 ± 28	514 ± 136	0.687

triple LP, |F|^(3/4)	115 ± 22	228 ± 19	416 ± 62	0.715
rho	1	1	1	0.242

- **Triple vs single and double large primes** at the same small base |F|^(2/3): triple is the cheapest, 2.0-3.5x below single and 1.1-1.3x below double.
- **Triple vs no large primes**: the lowest IC exponent is triple at |F|^(2/3) (0.687, against 0.721 with no large primes), but at these sizes its cost is higher (513 vs 379 x rho at p = 151). Triple at |F|^(3/4) is within noise of no large primes at every size.
- **Against rho**: every IC arm is 100-1,800x rho, and the ratio grows with p. The rho exponent here (0.24) is far below 1/2 because fixed setup dominates rho at n of 2^18-2^22, so exponent comparisons with rho at these sizes are not meaningful; the ratio column is.

Not done

- Quadruple large primes: no engine in the repository expresses them (prime-field script not written; Gaudry engine has 3-point relations).
- Prime-field k = 2, 3 (the main protocol): not run, for the same reason.
- Multi-target with k ≥ 2: not run.

Scripts: `scripts/mt.py`, `scripts/analyse_cubic.py`, `examples/klp_cubic.rs`; logs in `logs/`.